SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	**Total Marks:**	100
CO2 Statement:	*Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)*		
Student Name:	M Vishnu Vardhan	**Reg. No.:**	192472087
Section / Batch:	2024	**Date:**	27/07/2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

Q45. Linear Search – Negative Number Detection Given the unsorted dataset [12, -7, 25, 39, -14, 48, 56], apply Linear Search to find the element -14. Clearly interpret the problem and explain the search process step-by-step. Count the number of comparisons required before locating the target. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when searching for negative values in an unsorted list.

Code of Conduct

I __________________________(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __________________________

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Q45. Linear Search – Negative Number Detection
Dataset: [12, -7, 25, 39, -14, 48, 56] (indices 0 to 6)
Target: -14

Problem Interpretation
The dataset given is **unsorted**, which means there is no ordering property (like ascending/descending) that can be exploited to skip over elements. Because of this, **Linear Search** is the appropriate method — it examines each element one at a time, from the beginning of the array, comparing it against the target value (-14) until either a match is found or the entire array has been scanned.

Step-by-Step Search Process

Step	Index	Element	Comparison (Element vs -14)	Match?
1	0	12	$12 \neq -14$	No
2	1	-7	$-7 \neq -14$	No
3	2	25	$25 \neq -14$	No
4	3	39	$39 \neq -14$	No
5	4	-14	$-14 = -14$	**Yes ✓**

The search terminates at index 4 (5th element) since a match is found. No further elements (48, 56) need to be checked.
Comparisons Required
A total of **5 comparisons** were needed to locate the target -14, since it is positioned at index 4 (the 5th element in the array).

Time Complexity Analysis

Case	Description	Complexity
Best Case	Target is the very first element checked	O(1)
Average Case	Target is found somewhere in the middle on average (~n/2 comparisons)	O(n)
Worst Case	Target is the last element, or not present at all (full scan required)	O(n)

For this specific example (n = 7 elements, target found at position 5), the number of comparisons (5) is close to the array size, leaning toward the **average-to-worst case** range rather than the best case.

Verification of Correctness
Cross-checking the array directly: arr[4] = -14, which exactly matches the target value. Since no element before index 4 (12, -7, 25, 39) equals -14, the search correctly did not stop early, confirming that 5 comparisons were indeed the minimum necessary to find this specific target in this specific arrangement. The result is verified as **correct**.

Interpretation of Efficiency

- Linear Search operates **independently of the sign or magnitude** of the values being searched — it simply performs an equality check (element == target) at each position, whether the numbers are positive, negative, or zero.
- The presence of negative numbers (-7, -14) in the dataset has **no special effect** on the algorithm's behavior or efficiency; only the **position** of the target within the array determines how many comparisons are needed.
- Since the dataset is **unsorted**, more efficient methods like Binary Search cannot be applied without first sorting the data (which itself costs O(n log n)), making Linear Search the simplest and most direct choice here despite its O(n) worst-case time complexity.
- **Conclusion:** Linear Search is correct and reliable for this task, but it does not scale well — as the dataset grows larger, the number of comparisons grows proportionally (O(n)), making it inefficient for large-scale searches compared to methods that exploit sorted order or hashing.